

Lecture-32

Mathematics 1 (15B11MA111)

CO [C105.6]

Module: Matrices

Topic: Elementary Operations and Echelon Form

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Topics to be covered

-  Introduction to Matrices
-  Elementary Operations
-  Echelon Form
-  Solved Examples
-  Practice Questions

WHAT IS A MATRIX?

A matrix can be defined as a rectangular array or arrangement of numbers.

1. In other words, a set of 'mn' numbers arranged in the form of rectangular array of m rows and n columns is called $m \times n$ matrix read as m by n matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n}$$

2. The numbers $a_{11}, a_{12}, \dots$ etc are called elements of the matrix A.
3. Matrices are denoted by capital letters A, B, C etc.
4. A matrix is not a number. For example, 5 is just a number but [5] is a matrix having 1 row and 1 column.

TYPES OF MATRICES: QUICK REVISION

Type of Matrix	Properties	Example
Column Matrix	Number of columns =1 and m (finite) number of rows.	$A = \begin{bmatrix} 1 \\ 2 \\ 7 \\ 8 \end{bmatrix}_{4 \times 1}$
Row Matrix	Number of rows =1 and n (finite) number of columns.	$A = [1 \ 2 \ 3 \ 6 \ 12]_{1 \times 5}$
Rectangular Matrix	Number of rows and columns are unequal i.e. $m \neq n$.	$A = \begin{bmatrix} 2 & 5 & 6 \\ 1 & 0 & 5 \end{bmatrix}_{2 \times 3}$
Square Matrix	Number of rows and Columns are equal i.e. $m = n$. The elements 1,7,9 are known as diagonal elements and the line along which they lie is known as the PRINCIPAL DIAGONAL.	$A = \begin{bmatrix} 1 & 3 & 6 \\ 5 & 7 & 0 \\ 0 & 0 & 9 \end{bmatrix}_{3 \times 3}$

TYPES OF MATRICES: QUICK REVISION

Type of Matrix	Properties	Example
Diagonal Matrix	A <i>square</i> matrix in which all non-diagonal elements are zero	$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 7 \end{bmatrix}$
Scalar Matrix	A <i>diagonal</i> matrix in which diagonal elements are equal (but not equal to 1).	$\mathbf{A} = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix}$
Identity Matrix	A <i>square</i> matrix whose each diagonal element is unity and all other elements are zero.	$\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
Null (Zero) Matrix	A matrix of <i>any order</i> (rectangular or square) with every element as zero is called a null matrix (or a Zero matrix) and is denoted by O .	$\mathbf{O} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}; \mathbf{O} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

TYPES OF MATRICES: QUICK REVISION

Type of Matrix	Properties	Example
Upper Triangular Matrix	A <i>square matrix</i> is called an upper triangular matrix if all the elements below the principal diagonal are zero	$\mathbf{A} = \begin{bmatrix} 1 & 2 & 7 \\ 0 & 8 & 5 \\ 0 & 0 & 7 \end{bmatrix}$
Lower Triangular Matrix	A <i>square matrix</i> is called lower triangular matrix if all the elements above the principal diagonal are zero	$\mathbf{A} = \begin{bmatrix} 6 & 0 & 0 \\ 1 & 9 & 0 \\ 2 & 3 & 5 \end{bmatrix}$
Sub Matrix	A matrix obtained by deleting some rows or column or both of a given matrix is called its sub matrix.	If $\mathbf{A} = \begin{bmatrix} 1 & 5 & 0 \\ 9 & 1 & 7 \\ 8 & 0 & 1 \end{bmatrix}$
	If from matrix A, second row and third column is deleted then we get matrix B which is known as sub matrix of A. Every matrix is sub matrix of itself obtained by deleting no rows and no columns.	Then, sub matrix of A is B. $\mathbf{B} = \begin{bmatrix} 1 & 5 \\ 8 & 0 \end{bmatrix}$

ELEMENTARY OPERATIONS

- Elementary matrix operations (row or column) are ***simple operations*** that allow transforming a system of linear equations or a ***matrix*** into an ***equivalent system***, that is, into a new system of equations (or Matrix) having the same solutions as the original system.
- These operations play an important role in many matrix algebra applications, such as finding the inverse of a matrix and solving simultaneous linear equations.
- **Categories of Elementary Row Operations:**

1. Interchange of two rows or **Row switching**

A row within the matrix can be switched with another row i.e. $R_i \leftrightarrow R_j$.

2. Multiplication/division of any row by a non zero scalar or **Row multiplication**

Each element in a row can be multiplied by a non-zero constant i.e. $kR_i \rightarrow R_i, k \neq 0$.

3. Adding/Subtracting a scalar multiple of any row to another row or **Row Addition**

A row can be replaced by the sum of that row and a multiple of another row i.e.

$$R_i + kR_j \rightarrow R_i, i \neq j.$$

Note: Above three row operations if performed on columns instead of row are known as elementary column operations

Example: Consider the matrix $A = \begin{bmatrix} 2 & 5 \\ 1 & 6 \end{bmatrix}$. Apply the following operations on A,

(i) $R_1 \leftrightarrow R_2$;

(ii) $R_1 \leftarrow R_1 + 2R_2$, and

(iii) $R_1 \leftarrow 2R_1$.

Solution: (i) $A \approx \begin{bmatrix} 1 & 6 \\ 2 & 5 \end{bmatrix}$,

(ii) $A \approx \begin{bmatrix} 4 & 17 \\ 1 & 6 \end{bmatrix}$ and

(iii) $A \approx \begin{bmatrix} 4 & 10 \\ 1 & 6 \end{bmatrix}$

MINOR

Let $\mathbf{A}$ be an $m \times n$ matrix.

A minor of matrix $\mathbf{A}$ of order k is a determinant of a $k \times k$ sub-matrix of $\mathbf{A}$.

Or, Determinant of square sub matrix is termed as minor of matrix.

For example, determinant of sub matrix $\mathbf{B} = \begin{bmatrix} 1 & 5 \\ 8 & 0 \end{bmatrix}$ is minor of $\mathbf{A}$.

We can obtain the minors of order k from $\mathbf{A}$ by first deleting $m - k$ rows and $n - k$ columns, and then computing the determinant. There are usually many minors of $\mathbf{A}$ of a given order.

Example : Find the minors of order 3 of the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 2 & 4 & 2 \\ 0 & 2 & 2 & 1 \end{bmatrix}$$

Solution: The determinants of order 3 can be obtained by keeping all the rows and deleting one column from A. So there are four different minors of order 3.

One of them can be computed as:

$$\begin{vmatrix} 1 & 0 & 2 \\ 0 & 2 & 4 \\ 0 & 2 & 2 \end{vmatrix} = -4$$

The minors of order 3 are called the ***maximal minors of A***, since there are no 4×4 submatrices of A.

ECHELON FORM OF MATRIX

A $m \times n$ matrix is in **row echelon form** if the number of zeros preceeding the first non-zero entry of a row increases row by row until a row having all zero entries (or no other elimination is possible) is obtained. Therefore, a matrix is in row echelon form if it satisfies the following properties:

1. Rows containing all zeros occur only after all non-zero rows i.e. at the bottom.
2. If the i -th row contains all zeros, it is true for all subsequent rows.
3. If a column contains a non-zero entry of any row, then every subsequent entry in this column is zero.

Example:

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & 5 & 4 & 1 \\ 0 & 0 & 0 & 9 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 1 & -1 & 2 & 3 \\ 0 & 0 & 3 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Example: Reduce the given matrix to row-echelon form

$$\mathbf{A} = \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$$

Solution: Given $\mathbf{A} = \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$

Applying row operations, we have

$$\begin{aligned} &= \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & -21 & -14 & -29 \end{bmatrix} \begin{cases} R_2 \rightarrow R_2 + 2R_1 \\ R_3 \rightarrow R_3 - 7R_1 \end{cases} \\ &= \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & 0 & 0 & 0 \end{bmatrix} \{R_3 \rightarrow R_3 + 0.5R_2\} \end{aligned}$$

The last matrix is in row-echelon form and has two nonzero rows.

THANK YOU

Lecture-33

Mathematics 1 (15B11MA111)

CO [C105.6]

Module: Matrices

Topic: Rank of a Matrix

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Topics to be covered



Vectors



Rank of matrix



Solved Examples



Practice Questions

VECTORS

- A **vector** can be defined as a matrix with only one row or column.
- Its entries are called the **components** of the vector.
- Vectors are denoted by *lowercase* boldface letters **a**, **b**, ... or by its general component in brackets, $\mathbf{a} = [a_j]$, and so on.
- A (general) **row vector** is of the form $\mathbf{a} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n]$. Thus, in a matrix of order m by n, there are m row vectors and n column vectors.

- Similarly, a column vector is of the form $\mathbf{b} = \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \vdots \\ \mathbf{b}_m \end{bmatrix}$.

- A vector containing n components is known as n -tuple of numbers.

RANK OF A MATRIX

I. Rank of a matrix in terms of determinants

- The rank of a matrix $\mathbf{A}$, denoted by r or $r(\mathbf{A})$ is the order of the largest non-zero minor of $|\mathbf{A}|$.
- The rank of the matrix is the largest value of r , for which there exists at least one $r \times r$ sub matrix of $\mathbf{A}$ whose determinant is not zero.
- For an $m \times n$ matrix $r \leq \min(m, n)$.
- For a square matrix A of order n , the rank $r = n$ if $|A| \neq 0$, otherwise $r < n$.
- The rank of a matrix is 0 only if it is a null matrix.
- The rank of a matrix does not change if some elementary row operations are applied on matrix $\mathbf{A}$.

RANK OF A MATRIX

II. Rank of a matrix in terms of Echelon form

- In terms of row echelon form, the rank of a matrix A can be defined as the number of non-zero rows in the row echelon form of a matrix A .
- If A is a square matrix, then the row echelon form is an upper triangular matrix and the column echelon form is a lower triangular matrix.
- A non zero row implies having at least one element other than zero in that row.
- To find Rank of a matrix, try to reduce the given matrix to *row echelon form* by using *elementary row operations*.

Rank (A)= $r(A)$ = number of non-zero rows in A

RANK OF A MATRIX

III. Rank of a matrix in terms of vectors

The **rank** of a matrix is defined as:

- the maximum number of linearly independent *column* vectors in the matrix or, Or,
- the maximum number of linearly independent *row* vectors in the matrix.

Important points:

1. A set of m vectors each having n components is said to be **linearly dependent** if matrix formed using these vectors has **rank less than m** .
2. If **rank of those m vectors is equal to m** , then the vectors are **linearly independent**.
3. The maximum number of linearly independent vectors in a matrix is equal to the number of non-zero rows in its row echelon matrix.

Example: Determine rank of the matrix

$$\mathbf{A} = \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$$

Solution: Given $\mathbf{A} = \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$

Applying row operations, we have

$$\begin{aligned} &= \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & -21 & -14 & -29 \end{bmatrix} \left\{ \begin{array}{l} R_2 \rightarrow R_2 + 2R_1 \\ R_3 \rightarrow R_3 - 7R_1 \end{array} \right\} \\ &= \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & 0 & 0 & 0 \end{bmatrix} \{ R_3 \rightarrow R_3 + 0.5 R_2 \} \end{aligned}$$

The last matrix is in row-echelon form and has two nonzero rows.

Hence rank $\mathbf{A} = 2$.

Example: Reduce the matrix A to row echelon form and find its rank.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 5 \\ 1 & 5 & 5 & 7 \\ 8 & 1 & 14 & 17 \end{bmatrix}$$

Solution: Applying row operations:

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 5 \\ 1 & 5 & 5 & 7 \\ 8 & 1 & 14 & 17 \end{bmatrix} \left\{ \begin{array}{l} R_2 - 2R_1 \\ R_3 - R_1 \\ R_4 - 8R_1 \end{array} \right\}$$

$$\approx \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -2 & 5 \\ 0 & 3 & 2 & 3 \\ 0 & -15 & -10 & -15 \end{bmatrix} \left\{ \begin{array}{l} R_3 + R_2 \\ R_4 - 5R_2 \end{array} \right\}$$

$$\approx \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -2 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Rank (A)}=2$$

Example: Examine whether the set of vectors is linearly independent.

$$(1,2,3,4), (2,0,1,-2), (3,2,4,2).$$

Solution: Let each given vector represent a row of a matrix A. We reduce A to row-echelon form. If all the rows of the row echelon form have some non-zero elements, then the given set of vectors is linearly independent.

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 0 & 1 & -2 \\ 3 & 2 & 4 & 2 \end{bmatrix} \begin{cases} R_2 - 2R_1 \\ R_3 - 3R_1 \end{cases} \\ &\approx \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -4 & -5 & -10 \\ 0 & -4 & -5 & -10 \end{bmatrix} \{R_3 - R_2\} \\ &\approx \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -4 & -5 & -10 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Since, all the rows in the row echelon form of A are not non-zero, the given set of vectors are linearly dependent.

PRACTICE QUESTIONS:

1. Examine whether the set of vectors is linearly independent.

$(1,1,0,1), (1,1,1,1), (-1,1,1,1), (1,0,0,1).$

2. Reduce the matrix $\mathbf{A}$ to row echelon form and find its rank.

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 5 \\ 2 & -1 & 4 \\ -2 & 8 & 2 \end{bmatrix}$$

3. Find all values of μ for which the rank of the matrix $\mathbf{A}$ is 3.

$$\mathbf{A} = \begin{bmatrix} \mu & -1 & 0 & 0 \\ 0 & \mu & -1 & 0 \\ 0 & 0 & \mu & -1 \\ -6 & 11 & -6 & 1 \end{bmatrix}$$

THANK YOU

Lecture-34

Mathematics 1 (15B11MA111)

CO [C105.6]

Module: Matrices

Topic: Row Echleon form, Rank of a Matrix

Reference for the lecture

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Topics to be covered

-  **Row Echleon form**
-  **Rank of matrix**
-  **Linear Dependence and Independence of rows**
-  **Solved Examples**
-  **Practice Questions**

ECHELON FORM OF MATRIX

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3. If a column contains a non-zero entry of any row, then every subsequent entry in this column is zero.

Example:

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & 5 & 4 & 1 \\ 0 & 0 & 0 & 9 \end{bmatrix}$$

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Applying row operations, we have

$$\begin{aligned} &= \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & -21 & -14 & -29 \end{bmatrix} \left\{ \begin{array}{l} R_2 \rightarrow R_2 + 2R_1 \\ R_3 \rightarrow R_3 - 7R_1 \end{array} \right\} \\ &= \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & 0 & 0 & 0 \end{bmatrix} \left\{ R_3 \rightarrow R_3 + 0.5 R_2 \right\} \end{aligned}$$

The last matrix is in row-echelon form and has two nonzero rows.

RANK OF A MATRIX

Rank of a matrix in terms of Echelon form

- In terms of row echelon form, the rank of a matrix A can be defined as the number of non-zero rows in the row echelon form of a matrix A .
- If A is a square matrix, then the row echelon form is an upper triangular matrix and the column echelon form is a lower triangular matrix.
- A non zero row implies having at least one element other than zero in that row.
- To find Rank of a matrix, try to reduce the given matrix to *row echelon form* by using *elementary row operations*.

Rank (A)= $r(A)$ = number of non-zero rows in A

Example: Determine rank of the matrix

$$\mathbf{A} = \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$$

Solution: Given $\mathbf{A} = \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$

Applying row operations, we have

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The last matrix is in row-echelon form and has two nonzero rows.

Hence rank $\mathbf{A} = 2$.

Example: Reduce the matrix A to row echelon form and find its rank.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 5 \\ 1 & 5 & 5 & 7 \\ 8 & 1 & 14 & 17 \end{bmatrix}$$

Solution: Applying row operations:

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 5 \\ 1 & 5 & 5 & 7 \\ 8 & 1 & 14 & 17 \end{bmatrix} \left\{ \begin{array}{l} R_2 - 2R_1 \\ R_3 - R_1 \\ R_4 - 8R_1 \end{array} \right\}$$

$$\approx \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -2 & 5 \\ 0 & 3 & 2 & 3 \\ 0 & -15 & -10 & -15 \end{bmatrix} \left\{ \begin{array}{l} R_3 + R_2 \\ R_4 - 5R_2 \end{array} \right\}$$

$$\approx \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -2 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Rank (A)=2

LINEAR DEPENDENCE AND INDEPENDENCE OF ROWS OF A MATRIX

Linear dependence and independence of rows of a matrix

Using **rank** of a matrix we can find number of linear dependent/independent rows of a matrix as:

- maximum number of **linearly independent row vectors** in the matrix = rank of a matrix,
- number of **linearly dependent row vectors** in the matrix = number of rows of matrix – rank of matrix

Important points:

1. A set of m vectors each having n components is said to be **linearly dependent** if matrix formed using these vectors has **rank less than m** .
2. If **rank of those m vectors is equal to m** , then the vectors are **linearly independent**.
3. The maximum number of linearly independent vectors in a matrix is equal to the number of non-zero rows in its row echelon matrix.

Example: Examine whether the set of vectors is linearly independent.

$$(1,2,3,4), (2,0,1,-2), (3,2,4,2).$$

Solution: Let each given vector represent a row of a matrix A. We reduce A to row-echelon form. If all the rows of the row echelon form have some non-zero elements, then the given set of vectors is linearly independent.

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 0 & 1 & -2 \\ 3 & 2 & 4 & 2 \end{bmatrix} \left\{ \begin{array}{l} R_2 - 2R_1 \\ R_3 - 3R_1 \end{array} \right\} \\ &\approx \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -4 & -5 & -10 \\ 0 & -4 & -5 & -10 \end{bmatrix} \{ R_3 - R_2 \} \\ &\approx \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -4 & -5 & -10 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Since, all the rows in the row echelon form of A are not non-zero, the given set of vectors are linearly dependent.

PRACTICE QUESTIONS:

1. Examine whether the set of vectors is linearly independent.

$(1,1,0,1), (1,1,1,1), (-1,1,1,1), (1,0,0,1).$

2. Reduce the matrix $\mathbf{A}$ to row echelon form and find its rank.

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 5 \\ 2 & -1 & 4 \\ -2 & 8 & 2 \end{bmatrix}$$

3. Find all values of μ for which the rank of the matrix $\mathbf{A}$ is 3.

$$\mathbf{A} = \begin{bmatrix} \mu & -1 & 0 & 0 \\ 0 & \mu & -1 & 0 \\ 0 & 0 & \mu & -1 \\ -6 & 11 & -6 & 1 \end{bmatrix}$$

THANK YOU

Lecture-35

Mathematics 1 (15B11MA111)

CO [C105.6]

Module: Matrices

Topic: Homogenous System of Equations

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Homogeneous System of Linear Equations

A linear system of m equations in n unknowns $x_1, x_2, \dots, x_n$ is of the following form

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\&\vdots \\a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m\end{aligned}$$

If all b_i are zero then the system is known as **Homogeneous system** of linear equations.

Properties of Homogeneous System

Consider the homogeneous system of equations:

$$AX = 0$$

Where A is $m \times n$ matrix.

- The homogeneous system is always consistent since $x=0$ (trivial solution) is always a solution. So $\text{rank}(A) = \text{rank}(A|0)$.
- For nontrivial solution : $\text{rank}(A) < n$.
- **If $\text{rank}(A) = r < n \Rightarrow$** we obtain an $(n - r)$ parameter family of solutions which form a vector space of dimension $(n-r)$

- The solution space of homogeneous system is called the *null space* and its dimension is called the *nullity* of A .
- **If $\text{rank}(A) + \text{nullity}(A) = n$**

Remarks:

- **Remark 1:** If x_1 and x_2 are two solutions of a linear homogeneous system, then $\alpha x_1 + \beta x_2$ is also solution of the homogeneous system for any scalars α, β . This result does not hold for non-homogeneous systems.

- **Remark 2:** The homogeneous system of m equations in n unknowns and $m < n$, always possesses a non-trivial solution.
- **Remark 3:** If the non-homogeneous system of equations $AX = b$, Where A is $m \times n$ matrix ($m \geq n$) has a unique solution, i.e. $\text{rank}(A) = n$, then the corresponding homogeneous system $AX = 0$ has only the trivial solution, i.e. $x_h = 0$.

Examples

Example 1: Solve the following homogenous system of equation $AX = 0$ where A is given by

$$(i) \begin{bmatrix} 2 & 1 \\ 1 & -1 \\ 3 & 2 \end{bmatrix} \quad (ii) \begin{bmatrix} 1 & 2 & -3 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \quad (iii) \begin{bmatrix} 1 & 1 & -1 & 1 \\ 2 & 3 & 1 & 4 \\ 3 & 2 & -6 & 1 \end{bmatrix}$$

Find the rank (A) and nullity (A).

Solution:

$$(i) (A|0) = \left[\begin{array}{cc|c} 2 & 1 & 0 \\ 1 & -1 & 0 \\ 3 & 2 & 0 \end{array} \right] \begin{array}{l} R_2 - R_1/2 \\ R_3 - 3R_1/2 \end{array}$$
$$= \left[\begin{array}{cc|c} 2 & 1 & 0 \\ 0 & -3/2 & 0 \\ 0 & 1/2 & 0 \end{array} \right] R_3 + R_2/3$$

$$(A|0) = \left[\begin{array}{cc|c} 2 & 1 & 0 \\ 0 & -3/2 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

Since $\text{rank}(A) = 2 = \text{number of unknowns}$

Therefore the system has only a trivial solution. Hence, nullity $(A) = 0$.

Solution (ii):

$$(A|0) = \left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 1 & 1 & -1 & 0 \\ 1 & -1 & 1 & 0 \end{array} \right] \begin{array}{l} \\ R_2 - R_1 \\ R_3 - R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & -3 & 4 & 0 \end{array} \right] R_3 - 3R_2$$

$$(A|0) = \left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & -2 & 0 \end{array} \right]$$

Since rank (A)=3= number of unknowns

Therefore the system has only a trivial solution. Hence, nullity (A) = 0.

Solution (iii):

$$(A|0) = \left[\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 0 \\ 2 & 3 & 1 & 4 & 0 \\ 3 & 2 & -6 & 1 & 0 \end{array} \right] \begin{array}{l} \\ R_2 - 2R_1 \\ R_3 - 3R_1 \end{array}$$

$$= \left[\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 3 & 2 & 0 \\ 0 & -1 & -3 & -2 & 0 \end{array} \right] R_3 + R_2$$

$$(A|0) = \left[\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 3 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Since rank (A) =2 and the number of unknowns is 4. Hence we obtain a two parameter family of solutions as

$$x_2 = -3x_3 - 2x_4, \quad x_1 = -x_2 + x_3 - x_4 = 4x_3 + x_4,$$

Where, x_3, x_4 are arbitrary.

Therefore nullity (**A**) = 2.

THANK YOU

Lecture-35

Mathematics 1 (15B11MA111)

CO [C105.6]

Module: Matrices

Topic: Consistency of Non-Homogenous System of Equations

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Topics to be covered

-  **Linear System of Equations**
-  **Augmented Matrix**
-  **Solution of system of equations**
-  **Consistency of system of equations**
-  **Solved Examples**

LINEAR SYSTEM OF EQUATIONS

A linear system of m equations in n unknowns $x_1, x_2, \dots, x_n$ is of the following form

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\&\vdots \\a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m\end{aligned}$$

Note:

- If all b_j are zero, then the above system is called **Homogenous system**.
- If at least one b_j is non zero, then the above system is called **Non-Homogenous system**.

MATRIX FORM OF SYSTEM OF LINEAR EQUATIONS

Using Matrix multiplication, the linear system of ***m*** equations in ***n*** unknowns can be written as following vector equation

$$A\mathbf{X}=\mathbf{B}$$

$$\text{where, } A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \cdot & \cdot & \dots & \cdot \\ \cdot & \cdot & \dots & \cdot \\ \cdot & \cdot & \dots & \cdot \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}; \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix}$$

$$\text{and } \mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ \cdot \\ \cdot \\ \cdot \\ b_m \end{bmatrix}$$

AUGMENTED MATRIX

- **A** is a non-zero matrix of order $m \times n$, is called **Coefficient matrix**.
- **X** and **B** are column vectors with n and m component respectively.
- Augmented matrix $[A|B]$ obtained by appending the column **B** to the matrix **A**:

$$[A|B] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \cdot & \cdot & \dots & \cdot & \cdot \\ \cdot & \cdot & \dots & \cdot & \cdot \\ \cdot & \cdot & \dots & \cdot & \cdot \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{bmatrix}$$

SOLUTION OF LINEAR SYSTEM OF EQUATIONS

- **Solution:** A set of values (of variables) **satisfying** the given system of equations is termed as its solution.
- Let us consider a given system of equations in matrix form as $\mathbf{AX}=\mathbf{B}$.
- The coefficient matrix is given by $\mathbf{A}$ and corresponding Augmented matrix is represented as $[\mathbf{A}|\mathbf{B}]$.
- Evaluate the rank of $\mathbf{A}$ i.e. $r[\mathbf{A}]$ and rank of augmented matrix i.e. $r([\mathbf{A}|\mathbf{B}])$ using elementary row operations.

Now, the following cases arise:

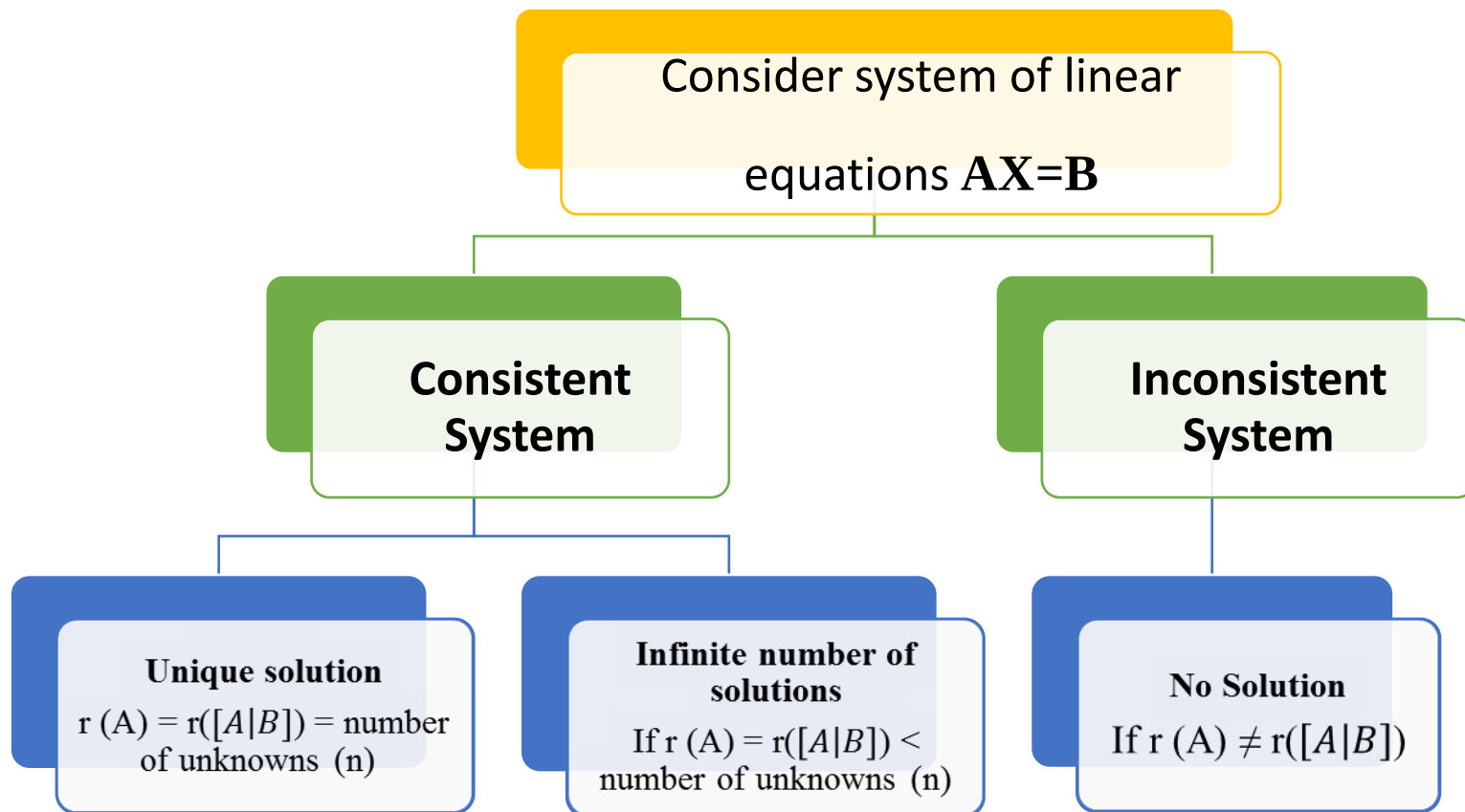
Unique Solution: If $r(\mathbf{A}) = r([\mathbf{A}|\mathbf{B}]) = \text{number of unknowns (n)}$

Infinite number of solutions: If $r(\mathbf{A}) = r([\mathbf{A}|\mathbf{B}]) < \text{number of unknowns (n)}$

No solution: If $r(\mathbf{A}) \neq r([\mathbf{A}|\mathbf{B}])$

CONSISTENCY OF THE SYSTEM AND SOLUTION

- ❖ **Consistent:** A system of linear equations is said to be consistent if it has **at least one solution**.
- ❖ **Inconsistent:** A system of linear equations is said to be inconsistent if it **has no solution**.



Example 1:

Test the consistency of the following system of linear equations

$$2x + y + z = 6$$

$$x + 3y + z = 6$$

$$x + y + 6z = 9$$

Solution:

Step 1: Convert the linear system of equations into matrix form $\mathbf{AX}=\mathbf{B}$.

$$\text{Here } A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 6 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B = \begin{bmatrix} 6 \\ 6 \\ 9 \end{bmatrix}$$

Step 2: Form the augmented matrix $[A|B]$.

$$[A|B] = \begin{bmatrix} 2 & 1 & 1 & 6 \\ 1 & 3 & 1 & 6 \\ 1 & 1 & 6 & 9 \end{bmatrix}$$

Step 3: Reduce $[A|B]$ using suitable elementary row transformations.

$$[A|B] = \begin{bmatrix} 2 & 1 & 1 & 6 \\ 1 & 3 & 1 & 6 \\ 1 & 1 & 6 & 9 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 1 & 6 & 9 \\ 1 & 3 & 1 & 6 \\ 2 & 1 & 1 & 6 \end{bmatrix} \quad R_3 \leftrightarrow R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 6 & 9 \\ 0 & 2 & -5 & -3 \\ 0 & -1 & -11 & -12 \end{bmatrix} \quad R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 6 & 9 \\ 0 & 2 & -5 & -3 \\ 0 & 0 & -27 & -27 \end{bmatrix} \quad R_3 \rightarrow 2R_3 + R_2$$

Row echelon form.

Step 4: Find the rank of A and rank of $[A|B]$:

Reduced row echelon form of $[A|B]$ is $\begin{bmatrix} 1 & 1 & 6 & 9 \\ 0 & 2 & -5 & -3 \\ 0 & 0 & -27 & -27 \end{bmatrix}$

$$r(A)=3$$

$$r([A|B])=3$$

Step 5: Check the consistency

Here number of unknowns =3

and $r(A) = r([A|B]) = \text{number of unknowns} = 3$

So system is consistent and possess unique solution.

Example 2:

Test the consistency of the following system of linear equations

$$2x + y + z = 6$$

$$x + 3y + z = 6$$

$$4x + 2y + 2z = 9$$

Solution:

Step 1: Convert the linear system of equations into matrix form $\mathbf{AX}=\mathbf{B}$.

$$\text{Here } A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 1 \\ 4 & 2 & 2 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad B = \begin{bmatrix} 6 \\ 6 \\ 9 \end{bmatrix}$$

Step 2: Form the augmented matrix $[A|B]$.

$$[A|B] = \begin{bmatrix} 2 & 1 & 1 & 6 \\ 1 & 3 & 1 & 6 \\ 4 & 2 & 2 & 9 \end{bmatrix}$$

Step 3: Reduce $[A|B]$ using suitable elementary row transformations.

$$[A|B] = \begin{bmatrix} 2 & 1 & 1 & 6 \\ 1 & 3 & 1 & 6 \\ 4 & 2 & 2 & 9 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 3 & 1 & 6 \\ 2 & 1 & 1 & 6 \\ 4 & 2 & 2 & 9 \end{bmatrix} \quad R_2 \leftrightarrow R_1$$

$$\sim \begin{bmatrix} 1 & 3 & 1 & 6 \\ 0 & -5 & -1 & -6 \\ 4 & 2 & 2 & 9 \end{bmatrix} \quad R_2 \rightarrow R_2 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 3 & 1 & 6 \\ 0 & -5 & -1 & -6 \\ 0 & -10 & -2 & -3 \end{bmatrix} \quad R_3 \rightarrow R_3 - 4R_1$$

$$\sim \begin{bmatrix} 1 & 1 & 6 & 9 \\ 0 & 2 & -5 & -3 \\ 0 & 0 & 0 & 9 \end{bmatrix} \quad R_3 \rightarrow R_3 - 2R_2$$

Row echelon form.

Step 4: Find the rank of A and rank of $[A|B]$:

Reduced row echelon form of $[A|B]$ is $\begin{bmatrix} 1 & 1 & 6 & 9 \\ 0 & 2 & -5 & -3 \\ 0 & 0 & 0 & 9 \end{bmatrix}$

$$r(A) = 2$$

$$r([A|B]) = 3$$

Step 5: Check the consistency

Here

$$r(A) \neq r([A|B])$$

So the system is **inconsistent**.

Example 3:

For what value of b and c following system of linear equations

$$\begin{aligned}x + 2y + z &= 7 \\ 3x + 2y + 2z &= 11 \\ 2x + 1y + bz &= c\end{aligned}$$

has

- I. No solution
- II. A unique solution
- III. Infinite number of solutions

Solution: augmented matrix $[A|B] = \begin{bmatrix} 1 & 2 & 1 & 7 \\ 3 & 2 & 2 & 11 \\ 2 & 1 & b & c \end{bmatrix}$

$$\sim \begin{bmatrix} 1 & 2 & 1 & 7 \\ 0 & -4 & -1 & -10 \\ 2 & 1 & b & c \end{bmatrix} \quad R_2 \rightarrow R_2 - 3R_1$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & 7 \\ 0 & -4 & -1 & -10 \\ 0 & -3 & b-2 & c-14 \end{bmatrix} \quad R_3 \rightarrow R_3 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & 7 \\ 0 & -4 & -1 & -10 \\ 0 & 0 & 4b-5 & 4c-26 \end{bmatrix} \quad R_3 \rightarrow 4R_3 - 3R_2$$

Row echelon matrix $\begin{bmatrix} 1 & 2 & 1 & 7 \\ 0 & -4 & -1 & -10 \\ 0 & 0 & 4b-5 & 4c-26 \end{bmatrix}$

Following three cases may arise

- I. No solution,
if $r(A) \neq r([A|B])$
i.e. when $4b - 5 = 0$ and $4c - 26 \neq 0$
- II. A unique solution,
if $r(A) = r([A|B]) = 3$ (no of unknowns)
i.e. when $4b - 5 \neq 0$
- III. Infinite number of solutions
if $r(A) = r([A|B]) < 3$ (no of unknowns)
 $4b - 5 = 0$ and $4c - 26 = 0$

THANK YOU

MATHEMATICS 1

(15B11MA111)

Lecture 36

Gauss Elimination method for system of linear equations

CONTENTS OF THE LECTURE

- Gauss elimination method
- Solution of system of homogenous equations
- Examples

GAUSS ELIMINATION METHOD

Step 1: Convert the linear system of equations into matrix form $AX=B$.

Step 2: Form the augmented matrix $[A|B]$.

Step 3: Forward elimination: $(n-1)$ number of steps are needed for forward elimination-

- a. Elimination of x_1 of from 2^{nd} to n^{th} equations
- b. Elimination of x_2 of from 3^{rd} to n^{th} equations
- c. Continue this process of elimination until n^{th} equation has only one non-zero coefficient (**Row Echelon form**).

Step 4: Check the consistency of the system.

Step 5: Backward substitution:

- i. Get the value of x_n from the n^{th} equation in Row Echelon form
- ii. Get the value of $x_{(n-1)}$ with the help of n^{th} and $(n-1)^{\text{th}}$ equations.
- iii. Continue this until the value of x_1 is obtained.

Example 1: (Unique Solution)

Solve the following system of linear equations; by Gauss elimination method

$$x_1 + x_2 + 6x_3 = 9$$

$$x_1 + 3x_2 + x_3 = 6$$

$$2x_1 + x_2 + x_3 = 6$$

Solution:

Form the augmented matrix $[A|B]$.

$$[A|B] = \begin{bmatrix} 1 & 1 & 6 & 9 \\ 1 & 3 & 1 & 6 \\ 2 & 1 & 1 & 6 \end{bmatrix}$$

Forward elimination :

$$[A|B] = \begin{bmatrix} 1 & 1 & 6 & 9 \\ 1 & 3 & 1 & 6 \\ 2 & 1 & 1 & 6 \end{bmatrix}$$

Elimination of x_1 of from 2nd to n^{th} equations –

$$\sim \begin{bmatrix} 1 & 1 & 6 & 9 \\ 0 & 2 & -5 & -3 \\ 0 & -1 & -11 & -12 \end{bmatrix} \quad R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - 2R_1$$

Elimination of x_2 of from 3rd to n^{th} equations –

$$\sim \begin{bmatrix} 1 & 1 & 6 & 9 \\ 0 & 2 & -5 & -3 \\ 0 & 0 & -27/2 & -27/2 \end{bmatrix} \quad R_3 \rightarrow R_3 + 1/2 R_2$$

Consistency:

Here $r(A) = r([A|B]) = \text{number of unknowns} = 3$

So the system is consistent and **possess unique solution.**

Backward substitution:

$$[A|B] \sim \begin{bmatrix} 1 & 1 & 6 & 9 \\ 0 & 2 & -5 & -3 \\ 0 & 0 & -27/2 & -27/2 \end{bmatrix}$$

$$-\frac{27}{2}x_3 = -\frac{27}{2} \Rightarrow x_3 = 1$$

$$2x_2 - 5x_3 = -3 \Rightarrow x_2 = 1$$

$$x_1 + x_2 + 6x_3 = 9 \Rightarrow x_1 = 2$$

So $x_1 = 2$, $x_2 = 1$, and $x_3 = 1$.

Example 2: (Infinite number of Solutions)

Solve the following system of linear equations; by Gauss elimination method

$$2x + y + z = 6$$

$$x + 3y + z = 6$$

$$3x + 4y + 2z = 12$$

Solution:

Step 1: Convert the linear system of equations into matrix form $\mathbf{AX}=\mathbf{B}$.

$$\text{Here } A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 1 \\ 3 & 4 & 2 \end{bmatrix}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad B = \begin{bmatrix} 6 \\ 6 \\ 12 \end{bmatrix}$$

Step 2: Form the augmented matrix $[A|B]$.

$$[A|B] = \begin{bmatrix} 2 & 1 & 1 & 6 \\ 1 & 3 & 1 & 6 \\ 3 & 4 & 2 & 12 \end{bmatrix}$$

Step 3: Forward elimination :

$$[A|B] = \begin{bmatrix} 2 & 1 & 1 & 6 \\ 1 & 3 & 1 & 6 \\ 3 & 4 & 2 & 12 \end{bmatrix}$$

Elimination of x_1 of from 2nd to n^{th} equations –

$$\sim \begin{bmatrix} 2 & 1 & 1 & 6 \\ 0 & \frac{5}{2} & \frac{1}{2} & 3 \\ 0 & \frac{5}{2} & \frac{1}{2} & 3 \end{bmatrix} \quad R_2 \rightarrow R_2 - \frac{1}{2}R_1 \text{ and } R_3 \rightarrow R_3 - \frac{3}{2}R_1$$

Elimination of x_2 of from 3rd to n^{th} equations –

$$\sim \begin{bmatrix} 2 & 1 & 1 & 6 \\ 0 & 5/2 & 1/2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_2$$

Consistency:

Here $r(A) = r([A|B]) = 2 < \text{number of unknowns} = 3$

So the system is consistent and **possess infinite number of solutions**

Backward substitution:

$$[A|B] \sim \begin{bmatrix} 2 & 1 & 1 & 6 \\ 0 & 5 & 1 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

x_3 is free variable

Choose $x_3 = k$

$$5/2 x_2 + 1/2 x_3 = 3 \Rightarrow x_2 = \frac{6 - k}{5}$$

$$2x_1 + x_2 + x_3 = 6 \Rightarrow x_1 = \frac{12 - 2k}{5}$$

So $x_1 = (12 - 2k)/5$, $x_2 = (6 - k)/5$, and $x_3 = k$.

Example 3: (Solution of system of homogenous equations)

Solve the following system of linear equations; by Gauss elimination method

$$\begin{aligned}x_1 + 3x_2 - 2x_3 &= 0 \\2x_1 - x_2 + 4x_3 &= 0 \\x_1 - 11x_2 + 14x_3 &= 0\end{aligned}$$

Solution:

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix}$$

Forward elimination :

Elimination of x_1 from 2nd to n^{th} equations –

$$A \sim \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & -14 & 16 \end{bmatrix} \quad R_2 \rightarrow R_2 - 2R_1 \text{ and } R_3 \rightarrow R_3 - R_1$$

Elimination of x_2 from 3rd to n^{th} equations –

$$A \sim \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & 0 & 0 \end{bmatrix} \quad R_3 \rightarrow R_3 - 2R_2$$

Consistency: Homogenous systems are always consistent

Here $r(A) < 3$

So the system **possess infinite number of solutions**

Backward substitution:

$$A \sim \begin{bmatrix} 1 & 3 & -2 \\ 0 & -7 & 8 \\ 0 & 0 & 0 \end{bmatrix}$$

x_3 is free variable

Choose $x_3 = k$

$$-7x_2 + 8x_3 = 0 \Rightarrow x_2 = \frac{8}{7}k$$

$$x_1 + 3x_2 - 2x_3 = 0 \Rightarrow x_1 = -\frac{10}{7}k$$

So $x_1 = -\frac{10}{7}k$, $x_2 = \frac{8}{7}k$, and $x_3 = k$.

Example 4: Solve the following system of linear equations; by Gauss elimination method

$$\begin{aligned}x_1 + x_2 + x_3 &= 4 \\2x_1 + 2x_2 + 3x_3 &= 9 \\3x_1 + 4x_2 - 2x_3 &= 9\end{aligned}$$

Solution:

Form the augmented matrix $[A|B]$.

$$[A|B] = \begin{bmatrix} 1 & 1 & 1 & 4 \\ 2 & 2 & 3 & 9 \\ 3 & 4 & -2 & 9 \end{bmatrix}$$

Forward elimination :

$$[A|B] = \begin{bmatrix} 1 & 1 & 1 & 4 \\ 2 & 2 & 3 & 9 \\ \textcolor{red}{3} & 4 & -2 & 9 \end{bmatrix}$$

$$R_3 \leftrightarrow R_1$$

$$[A|B] \sim \begin{bmatrix} 3 & 4 & -2 & 9 \\ 2 & 2 & 3 & 9 \\ 1 & 1 & 1 & 4 \end{bmatrix}$$

$$\sim \begin{bmatrix} 3 & 4 & -2 & 9 \\ 0 & \textcolor{red}{-2/3} & 13/3 & 3 \\ 0 & -1/3 & 5/3 & 1 \end{bmatrix} \quad R_2 \rightarrow R_2 - 2/3 R_1 \quad R_3 \rightarrow R_3 - 1/3 R_1$$

$$\sim \begin{bmatrix} 3 & 4 & -2 & 9 \\ 0 & -2/3 & 13/3 & 3 \\ 0 & 0 & -1/2 & -1/2 \end{bmatrix} \quad R_3 \rightarrow R_3 - 1/2 R_2$$

Consistency:

Here $r(A) = r([A|B]) = \text{number of unknowns} = 3$

So the system is consistent and **possess unique solution.**

Backward substitution:

$$[A|B] \sim \begin{bmatrix} 3 & 4 & -2 & 9 \\ 0 & -2/3 & 13/3 & 3 \\ 0 & 0 & -1/2 & -1/2 \end{bmatrix}$$

So $x_1 = 2$, $x_2 = 1$, and $x_3 = 1$

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Thank
you !

Lecture-36

Mathematics 1 (15B11MA111)

CO [C105.6]

Module: Matrices

Topic: Eigenvalues and eigenvectors

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Topic to be covered

- **Eigenvalues**
- **Eigenvectors**
- **Properties**
- **Cayley-Hamilton theorem**
- **Examples**

EIGEN VALUE PROBLEMS

Let $A = (a_{ij})$ be a square matrix of order n . the matrix A may be singular or non-singular. Let the homogeneous system of equations be

$$Ax = \lambda x \text{ or } (A - \lambda I)x = 0.$$

where λ is a scalar and I is the identity matrix of order n . The homogenous system of equations always has a trivial solution. We need to find values λ for which the homogeneous system has non-trivial solutions. The values of λ , for which non-trivial solutions of the homogeneous system exist, are called

the eigenvalues or characteristic values or latent values of A and the corresponding non-trivial solution vectors x are called the eigenvectors or the characteristic vectors of A .

If the homogeneous system has a non-trivial solution, then the rank of the coefficient matrix $(A - \lambda I)$ is less than n , that is, the coefficient matrix must be singular. Therefore,

$$\det(A - \lambda I) = 0.$$

Sum of eigenvalues = $\text{trace}(A)$

Product of eigenvalues = $|A|$

The set of eigenvalues is called **spectrum** of A .

Spectral radius of A = the largest eigenvalue in magnitude.

PROPERTIES OF EIGEN VALUES AND EIGEN VECTORS

Let λ be eigenvalue of A and x be its corresponding eigenvector. Then

- αA has eigenvalue $\alpha\lambda$ and the corresponding eigenvector is x .
- A^{-1} (*if exist*) has the eigenvalue $1/\lambda$.
- A and A^T has same eigenvalues.
- A^m has eigenvalue λ^m and the corresponding eigenvector is x for any positive integer m .

CAYLEY-HAMILTON THEOREM

Every square matrix A satisfies its own characteristic equation, that is

$$A^n - c_1 A^{n-1} + \cdots + (-1)^{n-1} c_{n-1} A + (-1)^n c_n I = 0$$

Example1: Find the eigenvalues and the eigenvectors of

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}.$$

Solution: The characteristic equation is given as

$$\det(A - \lambda I) = 0, \begin{vmatrix} 1 - \lambda & 0 & 0 \\ 0 & 2 - \lambda & 1 \\ 2 & 0 & 3 - \lambda \end{vmatrix} = 0$$

$$(1 - \lambda)(2 - \lambda)(3 - \lambda) = 0, \quad \lambda = 1, 2, 3$$

Corresponding to the eigenvalue $\lambda=1$, we have

$$(A - I)x = 0 \text{ or } (A - I)x=0$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ or } \begin{cases} x_2 + x_3 = 0 \\ x_1 + x_3 = 0 \end{cases}$$

Choosing x_3 arbitrary that is $x_3 = 1$, we get eigenvector as $[1, -1, 1]^T$.

Corresponding to the eigenvalue $\lambda=2$, we have

$$(A - 2I)x = 0 \text{ or } (A - 2I)x=0$$

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

or $x_1 = 0$, $x_3 = 0$ and x_2 arbitrary. Taking $x_2 = 1$ we get eigenvector as $[0, 1, 0]^T$.

Corresponding to the eigenvalue $\lambda=3$, we have

$$(A - I)x = 0 \text{ or } (A - 3I)x = 0$$
$$\begin{bmatrix} -2 & 0 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ or } \begin{cases} x_1 = 0 \\ x_2 + x_3 = 0 \end{cases}$$

or Taking $x_3 = 1$ we get eigenvector as $[0, -1, 1]^T$.

Example2: Find the eigenvalues and the

eigenvectors of $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

Solution: The characteristic equation is given as

$$\det(A - \lambda I) = 0, \begin{vmatrix} 1 - \lambda & 1 & 0 \\ 0 & 1 - \lambda & 1 \\ 0 & 0 & 1 - \lambda \end{vmatrix} = 0$$

$$(1 - \lambda)(1 - \lambda)(1 - \lambda) = 0, \quad \lambda = 1, 1, 1$$

Eigenvector corresponding to the eigenvalue $\lambda = 1$, we have

$$(A - I)x = 0 \text{ or } (A - I)x = 0$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ or } \begin{cases} x_2 = 0 \\ x_3 = 0 \\ x_1 \text{ arbitrary} \end{cases}$$

or choosing $x_1 = 1$ we get eigenvector as $[1, 0, 0]^T$.
Hence A has only one independent eigenvector.

PRACTICE QUESTIONS:

1. Find the eigenvalues and the eigenvectors of the following

$$(i) A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(ii) A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2. Find the spectral radius of $A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$

THANK YOU

Lecture-39

Mathematics 1 (15B11MA111)

CO [C105.6]

Module: Matrices

Topic: Reduction to diagonal form

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Topics to be covered



Similarity Transformation



Diagonalization of a matrix



Criterion for Diagonalizability



Solved Examples



Practice Questions

Similarity Transformation

- **Similar Matrices**: Let A and B be square matrices of the same order. The matrix A is said to be similar to the matrix B if there exists an invertible matrix P such that

$$A = P^{-1} B P$$

The matrix P is called the similarity matrix. The transformation is called a similarity transformation.

Theorem: Similar matrices have the same characteristic equation (and hence the same eigen values). Further, if x is an eigenvector of A corresponding to the eigenvalue λ , then $P^{-1} x$ is an eigenvector of B corresponding to the eigenvalue λ , where P is the similarity matrix.

Diagonalization of a matrix

- Diagonalisation of a matrix A is the process of reduction of A to a diagonal form.

If A is related to D by a similarity transformation such that $D = P^{-1} A P$ then A is reduced to the diagonal matrix D through model matrix P . D is also called **spectral matrix** of A .

OR

If an $n \times n$ nonsingular matrix P can be found so that $P^{-1} A P = D$ is a diagonal matrix, then we say that the $n \times n$ matrix A can be diagonalized, or its diagonalizable and that P diagonalizes A .

Criterion for Diagonalizability

Theorem: A square matrix A of order n is diagonalizable if and only if it has n linearly independent eigenvectors.

Sufficient condition for Diagonalizability

- **Theorem:** A square matrix A of order n has distinct eigenvalues, it is diagonalizable.
- **Note:** If the matrix A does not have n distinct eigenvalues, then it may or may not be diagonalizable.

- Notes**
1. The square matrix P , which diagonalises A , is found by grouping the eigen vectors of A into square-matrix and the resulting diagonal matrix has the eigen values of A as its diagonal elements.
 2. The transformation of a matrix A to $P^{-1} A P$ is known as a *similarity transformation*.
 3. The reduction of A to a diagonal matrix is, obviously, a particular case of similarity transformation.
 4. The matrix P which diagonalises A is called the *modal matrix* of A and the resulting diagonal matrix D is known as the *spectra matrix* of A .

POWERS OF A MATRIX (By diagonalisation)

We can obtain powers of a matrix by using diagonalisation.

We know that $D = P^{-1} A P$

Where A is the square matrix and P is a non-singular matrix.

$$D^2 = (P^{-1} A P) (P^{-1} A P) = P^{-1} A (P P^{-1}) A P = P^{-1} A^2 P$$

Similarly $D^3 = P^{-1} A^3 P$

In general $D^n = P^{-1} A^n P$

Pre-multiply (1) by P and post-multiply by P^{-1}

$$\begin{aligned} P D^n P^{-1} &= P (P^{-1} A^n P) P^{-1} \\ &= (P P^{-1}) A^n (P P^{-1}) \\ &= A^n \end{aligned}$$

Procedure: (1) Find eigen values for a square matrix A .

(2) Find eigen vectors to get the modal matrix P .

(3) Find the diagonal matrix D , by the formula $D = P^{-1} A P$

(4) Obtain A^n by the formula $A^n = P D^n P^{-1}$.

- **Example1** Let $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ Find matrix P such that $P^{-1}AP$ is diagonal matrix.

Solution. The characteristic equation of the matrix A is

$$\begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (6-\lambda)[9+\lambda^2-6\lambda-1]+2[-6+2\lambda+2]+2[2-6+2\lambda]=0$$

$$\Rightarrow (6-\lambda)(\lambda^2-6\lambda+8)-8+4\lambda-8+4\lambda=0$$

$$\Rightarrow 6\lambda^2-36\lambda+48-\lambda^3+6\lambda^2-8\lambda-16+8\lambda=0$$

$$\Rightarrow -\lambda^3+12\lambda^2-36\lambda+32=0 \quad \Rightarrow \quad \lambda^3-12\lambda^2+36\lambda-32=0$$

$$\Rightarrow (\lambda-2)^2(\lambda-8)=0 \quad \Rightarrow \quad \lambda = 2, 2, 8$$

Eigen vector for $\lambda = 2$

$$\begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 2 & -1 & 1 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \begin{matrix} R_2 \rightarrow R_1 + R_2 \\ R_3 \rightarrow R_2 + R_3 \end{matrix}$$

$$\begin{bmatrix} 2 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{or} \quad 2x_1 - x_2 + x_3 = 0$$

This equation is satisfied by $x_1 = 0, x_2 = 1, x_3 = 1$

$$X_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

and again

$$x_1 = 1, x_2 = 3, x_3 = 1.$$

$$X_2 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

Eigen vector for $\lambda = 8$

$$\begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-2x_1 - 2x_2 + 2x_3 = 0$$

$$-2x_1 - 5x_2 - x_3 = 0$$

$$\frac{x_1}{2+10} = \frac{x_2}{-4-2} = \frac{x_3}{10-4} \Rightarrow \frac{x_1}{12} = \frac{x_2}{-6} = \frac{x_3}{6} \Rightarrow \frac{x_1}{2} = \frac{x_2}{-1} = \frac{x_3}{1}$$

$$X_3 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$\therefore P = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 3 & -1 \\ 1 & 1 & 1 \end{bmatrix}, \quad P^{-1} = -\frac{1}{6} \begin{bmatrix} 4 & 1 & -7 \\ -2 & -2 & 2 \\ -2 & 1 & -1 \end{bmatrix}$$

$$\text{Now } P^{-1}AP = -\frac{1}{6} \begin{bmatrix} 4 & 1 & -7 \\ -2 & -2 & 2 \\ -2 & 1 & -1 \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \\ 1 & 3 & -1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

Example 2: Diagonalize the matrix $\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{pmatrix}$.

Solution: eigenvalues and corresponding eigenvectors are

$$\lambda_1 = 0, \quad \lambda_2 = -4, \quad \lambda_3 = 3, \quad \mathbf{K}_1 = \begin{pmatrix} 1 \\ 6 \\ -13 \end{pmatrix}, \quad \mathbf{K}_2 = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{K}_3 = \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix}.$$

Since the eigenvalues are distinct, $\mathbf{A}$ is diagonalizable. We form the matrix

$$\mathbf{P} = (\mathbf{K}_1 \mathbf{K}_2 \mathbf{K}_3) = \begin{pmatrix} 1 & -1 & 2 \\ 6 & 2 & 3 \\ -13 & 1 & -2 \end{pmatrix}.$$

Matching the eigenvalues with the order in which the eigenvectors appear in $\mathbf{P}$, we know that the diagonal matrix will be

$$\mathbf{D} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Now from either of the methods of Section 8.6 we find

$$\mathbf{P}^{-1} = \begin{pmatrix} -\frac{1}{12} & 0 & -\frac{1}{12} \\ -\frac{9}{28} & \frac{2}{7} & \frac{3}{28} \\ \frac{8}{21} & \frac{1}{7} & \frac{2}{21} \end{pmatrix},$$

and so

$$\begin{aligned} \mathbf{P}^{-1}\mathbf{A}\mathbf{P} &= \begin{pmatrix} -\frac{1}{12} & 0 & -\frac{1}{12} \\ -\frac{9}{28} & \frac{2}{7} & \frac{3}{28} \\ \frac{8}{21} & \frac{1}{7} & \frac{2}{21} \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 2 \\ 6 & 2 & 3 \\ -13 & 1 & -2 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 3 \end{pmatrix} = \mathbf{D}. \end{aligned}$$

PRACTICE QUESTIONS:

1. Find the matrix B which transforms the matrix

$$A = \begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix} \text{ to a diagonal matrix.}$$

$$\text{Ans. } B = \begin{bmatrix} 4 & 3 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$$

2. For the matrix $A = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{bmatrix}$, determine a matrix P such that $P^{-1}AP$ is diagonal matrix.

$$\text{Ans. } P = \begin{bmatrix} -1 & 1 & 1 \\ 0 & -\sqrt{2} & \sqrt{2} \\ 1 & 1 & 1 \end{bmatrix}$$

3. Determine the eigen values and the corresponding eigen vectors of the matrix $A = \begin{bmatrix} 5 & 7 & -5 \\ 0 & 4 & -1 \\ 2 & 8 & -3 \end{bmatrix}$

Hence find the matrix P such that $P^{-1}AP$ is diagonal matrix.

$$\text{Ans. } P = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 1 & 1 \\ 3 & 2 & 1 \end{bmatrix}$$

THANK YOU

Lecture- 40
Mathematics 1
(15B11MA111)
Module: Matrices
CO [C105.4]

Topics to be covered

- Quadratic forms
- Examples
- Practice Questions
- Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Quadratic forms

A **quadratic form** in n independent variables $(x_1, x_2, x_3, \dots, x_n)$ is a homogeneous second degree polynomial of the form

$$\begin{aligned} V(x_1, x_2, \dots, x_n) &= \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j \\ &= a_{11}x_1^2 + a_{12}x_1x_2 + \dots + a_{1n}x_1x_n + a_{21}x_2^2 + a_{21}x_2x_1 + \dots + \\ &\quad a_{2n}x_2x_n + \dots + a_{n1}x_1^2 + a_{n2}x_1x_2 + \dots + a_{nn}x_nx_n \end{aligned} \quad (1)$$

Defining the vector $x = [x_1 \quad x_2 \quad \dots \quad x_n]^T$ and the matrix

$$A = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & \ddots & \vdots \\ \vdots & & & \ddots \end{bmatrix}$$

The quadratic form (1) may be written in the form

$$\underline{\hspace{1cm}} V(x) = x^T A x \tag{2}$$

The matrix A is referred to as the matrix of the quadratic form and the determinant of A is called the discriminant of the quadratic form.

Example: Find the real symmetric matrix corresponding to the quadratic form

$$V(x_1, x_2, x_3) = x_1^2 + 3x_2^2 - 4x_3^2 - 3x_1x_2 + 2x_1x_3 - 5x_2x_3$$

Solution: If $x = [x_1 \quad x_2 \quad x_3]^T$, we have

$$V(x_1, x_2, x_3) = [x_1 \quad x_2 \quad x_3] \begin{bmatrix} 1 & -\frac{3}{2} & 1 \\ -\frac{3}{2} & 3 & -\frac{5}{2} \\ 1 & -\frac{5}{2} & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x^T A x$$

where the matrix of the quadratic form is

$$A = \begin{bmatrix} 1 & -\frac{3}{2} & 1 \\ -\frac{3}{2} & 3 & -\frac{5}{2} \\ 1 & -\frac{5}{2} & -4 \end{bmatrix}$$

Since, a **real symmetric matrix, A can always be diagnosable.** So,

$$\hat{M}^T A \hat{M} = B$$

where $\hat{M}$ is the normalized orthogonal modal matrix of A and B is its special matrix.
Thus for a real quadratic form we can specify a change of variables

$$x = \hat{M}y$$

where $y = [y_1 \quad y_2 \quad \dots \quad y_n]^T$ such that $V = x^T A x = y^T \hat{M}^T A \hat{M} y = y^T B y$
giving $V = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2$ (3)

The resulting form of V is called **canonical form.**

Example: Find the canonical form of the quadratic form

$$V = 2x_1^2 + 5x_2^2 + 3x_3^2 + 4x_1x_2$$

Can V take any negative values for any value of x_1, x_2 and x_3 ?

Solution: We have $V = x^T \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix} x = x^T A x$

where $x = [x_1 \quad x_2 \quad x_3]^T$, $A = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

$$\hat{M} = \begin{bmatrix} \sqrt{\frac{1}{5}} & 0 & -2\sqrt{\frac{1}{5}} \\ 2\sqrt{\frac{1}{5}} & 0 & \sqrt{\frac{1}{5}} \\ 0 & 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

such that $\hat{M}^T A \hat{M} = B$. Thus setting $x = \hat{M}y$, we obtain

$$V = y^T \hat{M}^T A \hat{M} y = y^T \begin{bmatrix} 6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} y = 6y_1^2 + 3y_2^2 + y_3^2$$

as the required canonical form.

Clearly V is non-negative for all y_1, y_2 and y_3 . Since $x = \hat{M}y$ and $\hat{M}$ is an orthogonal matrix it follows that $y = \hat{M}^T x$, so for all x there is a corresponding y . It follows that V cannot take negative values for any values of x_1, x_2 and x_3 , such a quadratic form $x^T A x$ is called a **positive-definite quadratic form**. This property depends only on the eigen value of real symmetric matrix **A**.

Classification of Quadratic Forms:

- (a) V is **positive definite**, that is $V > 0$ for all vectors x except $x=0$, if and only if **all the eigenvalues of A are positive**.
- (B) V is **positive semi definite**, that is $V \geq 0$ for all vectors x and $V=0$ for at least one vector $x \neq 0$, if and only if **all the eigenvalues of A are non – negative and at least one of the eigenvalues is zero**.
- (c) V is **negative definite** if $-V$ is **positive definite**, with a corresponding condition on the **eigenvalues of $-A$** .
- (d) V is **negative – semidefinite** if $-V$ is **positive semidefinite**, with a corresponding condition on the **eigenvalues of $-A$** .
- (e) V is **indefinite**, that is V takes at least **one positive value and at least one negative value**, if and only if the matrix A has both **positive and negative eigenvalues**.

Example

Classify the following quadratic forms:

(a) $3x_1^2 + 2x_2^2 + 3x_3^2 - 2x_1x_2 - 2x_2x_3$

(b) $7x_1^2 + x_2^2 + x_3^2 - 4x_1x_2 - 4x_1x_3 + 8x_2x_3$

(c) $-3x_1^2 - 5x_2^2 - 3x_3^2 + 2x_1x_2 + 2x_2x_3 - 2x_1x_3$

(d) $4x_1^2 + x_2^2 + 15x_3^2 - 4x_1x_2$

Solution

(A) The matrix corresponding to the quadratic form is

$$A = \begin{bmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}$$

The eigenvalues of A are 4, 3 and 1, so the quadratic form is **positive definite**.

(b) The matrix corresponding to the quadratic form is

$$A = \begin{bmatrix} 7 & -2 & -2 \\ -2 & 1 & 4 \\ -2 & 4 & 1 \end{bmatrix}$$

(c) The matrix corresponding to the quadratic form is

$$A = \begin{bmatrix} 3 & 1 & -1 \\ 1 & -5 & 1 \\ -1 & 1 & -3 \end{bmatrix}$$

The eigenvalues of A are -6, -3 and -2, so the quadratic form is **negative-definite**.

(d) The matrix corresponding to the quadratic form is

$$A = \begin{bmatrix} 4 & -2 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 15 \end{bmatrix}$$

The eigenvalues of A are 15, 5 and 0, so the quadratic form is **positive semi-definite**.

Assignment

1. Show that the matrix $A = \begin{bmatrix} 2 & 1 & 3 \\ -3 & 4 & -1 \\ -1 & 1 & 2 \end{bmatrix}$ is positive definite.
2. Find the symmetric matrix A for the quadratic form given as:
 - (i) $x_1^2 - x_2^2 + x_3^2 - 2x_1x_2 + 4x_2x_3$.
 - (ii) $x_1^2 + x_2^2 + 3x_3^2 - x_1x_2 - 2x_2x_3 - 4x_1x_3$.

Thank You